MODULE 46 · DOMAIN 5 - OPERATIONAL EXCELLENCE AND AUTOMATION

User Experience Monitoring and SLOs

Measure what users actually experience, set objectives for it, alert on how fast the error budget is burning, and start the fix automatically.

DOMAIN 5 of 5 EXAM WEIGHT 20% MODULE 46 of 52

admin@sthithapragna.com

Measure the user's experience, judge it against an objective, and alert on the budget burning, not on every blip.

admin@sthithapragna.com

Watching from the user's side

TOOL	MEASURES	BEST AT
CloudWatch Synthetics	Scripted canaries on a schedule	Finding breakage before users do
CloudWatch RUM	Real browser and mobile sessions	Actual load times and errors by geography
Application Signals	Service operations and dependencies	Latency, faults and SLOs per service
Internet Monitor	Internet paths to your application	Provider and location problems

ON THE EXAM

Synthetics works with no users; RUM needs them.

Synthetics or RUM?

SYNTHETICS CANARIES

Simulated users

- Runs every few minutes, any hour
- Scripted journeys, screenshots
- Catches breakage at 3 a.m.
- Same path every time

REAL USER MONITORING

Actual users

- Every real session
- Browsers, devices, countries
- Web vitals and JavaScript errors
- Silent when nobody visits

THE TIEBREAKER

Know before users notice, at quiet times: Synthetics. What real users see, by place and device: RUM.

SLO building blocks

TERM	MEANING
SLI	The measure: availability or latency
Attainment goal	For example 99.9% over 28 days
Error budget	The failure allowed by the goal
Burn rate	How fast the budget is being spent
Period-based	Each minute healthy or unhealthy
Request-based	Good requests over total requests

ON THE EXAM

Burn rate above 1 means the goal will be missed if nothing changes.

Burn-rate alerting

01
Long window
For example 1 hour, high burn

02
Short window
One twelfth: 5 minutes

03
Composite alarm
Both in alarm together

04
Page
Only while it is still happening

ON THE EXAM

Pair a long and a short window so pages fire for real, ongoing budget loss.

From alarm to action

ALARM ACTION	RESULT
SNS topic	Notify people or systems
Systems Manager OpsItem	Track and review in OpsCenter
EventBridge rule	Start an SSM Automation runbook
Auto Scaling or EC2 action	Scale, stop, reboot or recover
Lambda function	Run custom remediation

ON THE EXAM

Known fix for a known alarm: run it automatically with SSM Automation.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Broken at night, found by customers	Synthetics canary
Slow for some users in some countries	CloudWatch RUM
Alert on objective, not every spike	SLO with burn-rate alarms
Maintenance counts against the SLO	Exclusion windows
Low traffic, one failure spoils a minute	Request-based SLI
Known fix, run it automatically	Alarm to SSM Automation

ON THE EXAM

Measure the user, judge against the objective, act automatically.

> Ashgrove Travel's booking flow broke at 03:00 after a dependency change. Traffic is light at night, and customers reported it at 08:00.
>
> The team wants the full booking journey checked every five minutes, around the clock, with an alarm on failure.

What should the solutions architect implement?

A CloudWatch RUM on the booking pages

B A CloudWatch Synthetics canary that scripts the booking journey, with an alarm on its success metric

C A Route 53 health check on the home page

D A metric filter on application error logs

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A RUM

Needs real users; there were almost none.

B Synthetics canary

Runs the journey every five minutes, users or not.

C Health check

Home page may load while booking fails.

D Log metric filter

Only sees errors the app logs, and needs traffic.

THE SIGNAL

Catch breakage when nobody is using the site: a Synthetics canary.

> Ashgrove Travel's canaries in eu-west-1 show fast page loads, but customers in Brazil and India complain that the site is slow on mobile.
>
> The team wants actual page-load times and JavaScript errors, broken down by country, browser and device.

What should the solutions architect add?

A More canaries in eu-west-1

B X-Ray tracing on the back end

C CloudWatch RUM, with its web client added to the site

D Application Load Balancer access logs

◆ PAUSE AND DECIDE

What should the solutions architect add?

A More canaries

Same Region, same browser; not real users.

B X-Ray

Back-end timing; misses the browser.

C RUM

Real sessions by country, browser and device.

D ALB logs

Server timing; no page rendering or JS errors.

THE SIGNAL

What real users see, by place and device: RUM.

> Ashgrove Travel promises 99.9% availability for its search API over a rolling 28 days. A static alarm on any 5xx minute pages the on-call engineer several times a night for brief spikes.
>
> The team wants pages only when the promise is genuinely at risk.

What should the solutions architect implement?

A. An Application Signals SLO with burn-rate alarms on a long and a short window, joined in a composite alarm

B. Raise the static threshold until the pages stop

C. Anomaly detection on the 5xx count

D. A dashboard reviewed each morning

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A. SLO burn-rate alarms
Pages when the budget is genuinely burning.

B. Higher threshold
Arbitrary; may hide real outages.

C. Anomaly detection
Unusual is not the same as at risk.

D. Dashboard
No alerting at all.

THE SIGNAL

Alert when the promise is at risk, not on every spike: SLO burn-rate alarms.

> Ashgrove Travel takes the search API down for planned maintenance every Sunday from 02:00 to 02:30, with customers notified in advance.
>
> Those 30 minutes consume the SLO error budget every week, and the agreement with customers excludes them.

What should the solutions architect do?

A Lower the attainment goal

B Delete and recreate the SLO after each window

C Disable the SLO alarms by hand on Sundays

D Add a recurring exclusion window to the SLO for the maintenance period

◆ PAUSE AND DECIDE

What should the solutions architect do?

A Lower goal

Misstates the promise every other day.

B Recreate

Loses history; absurd effort.

C Manual disable

Budget still consumed; relies on memory.

D Recurring exclusion window

Maintenance excluded from measurement.

THE SIGNAL

Planned maintenance outside the agreement: SLO exclusion windows.

> Ashgrove Travel's partner API receives about 20 requests per minute overnight. With a period-based availability SLO, one failed request marks a whole minute unhealthy.
>
> Overnight attainment looks far worse than the share of requests that actually failed.

What should the solutions architect change?

A Use a 5-minute period

B Switch to a latency SLI

C Use a request-based SLI that counts good requests against total requests

D Exclude the night from the SLO

◆ PAUSE AND DECIDE

What should the solutions architect change?

A Longer period

One failure now spoils five minutes.

B Latency SLI

Measures something else.

C Request-based

Attainment reflects the real failure ratio.

D Exclude nights

Hides real failures from the objective.

THE SIGNAL

Low or uneven traffic, period health misleading: request-based SLI.

> Ashgrove Travel's EC2 batch servers regularly fill their temporary volume. An engineer is paged, logs in, runs the same cleanup commands, and closes the alert.
>
> The team wants the cleanup to run automatically, with a record of each occurrence for review.

What should the solutions architect implement?

A. An SNS email to the whole team

B. A disk-usage alarm that creates a Systems Manager OpsItem, and an EventBridge rule on the alarm that starts an SSM Automation runbook to clean up

C. Replace the instances every night

D. Double the volume size

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Team email

Still manual, and noisier.

B OpsItem plus Automation

Fix runs itself; each occurrence recorded.

C Nightly replacement

Disrupts batch jobs; ignores the trigger.

D Bigger volume

Delays the same problem.

THE SIGNAL

Known fix, run automatically, with a record: alarm to SSM Automation and OpsItem.

What to carry out of this module

- **Synthetics** before users notice; **RUM** for what real users see.
- **SLOs** in Application Signals: SLI, goal, error budget.
- **Burn-rate alarms**, long and short window, composite.
- **Exclusion windows**; **request-based** for low traffic.
- **Alarm to SSM Automation** for known fixes.

> ON THE EXAM
>
> Measure the user, judge against the objective, act automatically.